



किसान CREDIT

**FROM FARM GPS TO CARBON CREDIT PAYOUT –
A FULLY AUTOMATED, TRANSPARENT,
AND BUILT FOR SMALLHOLDERS.**



THE PROBLEM

- Global carbon credit market is worth ₹2.5 trillion+, but small farmers in Nagpur earn almost zero from it.
- Current verification is manual, slow, and expensive requires scientists to visit fields, paperwork, and middlemen.

THE RESULT

Small orange and cotton farmers cannot prove their climate impact, so buyers don't trust their claims and farmers lose income.

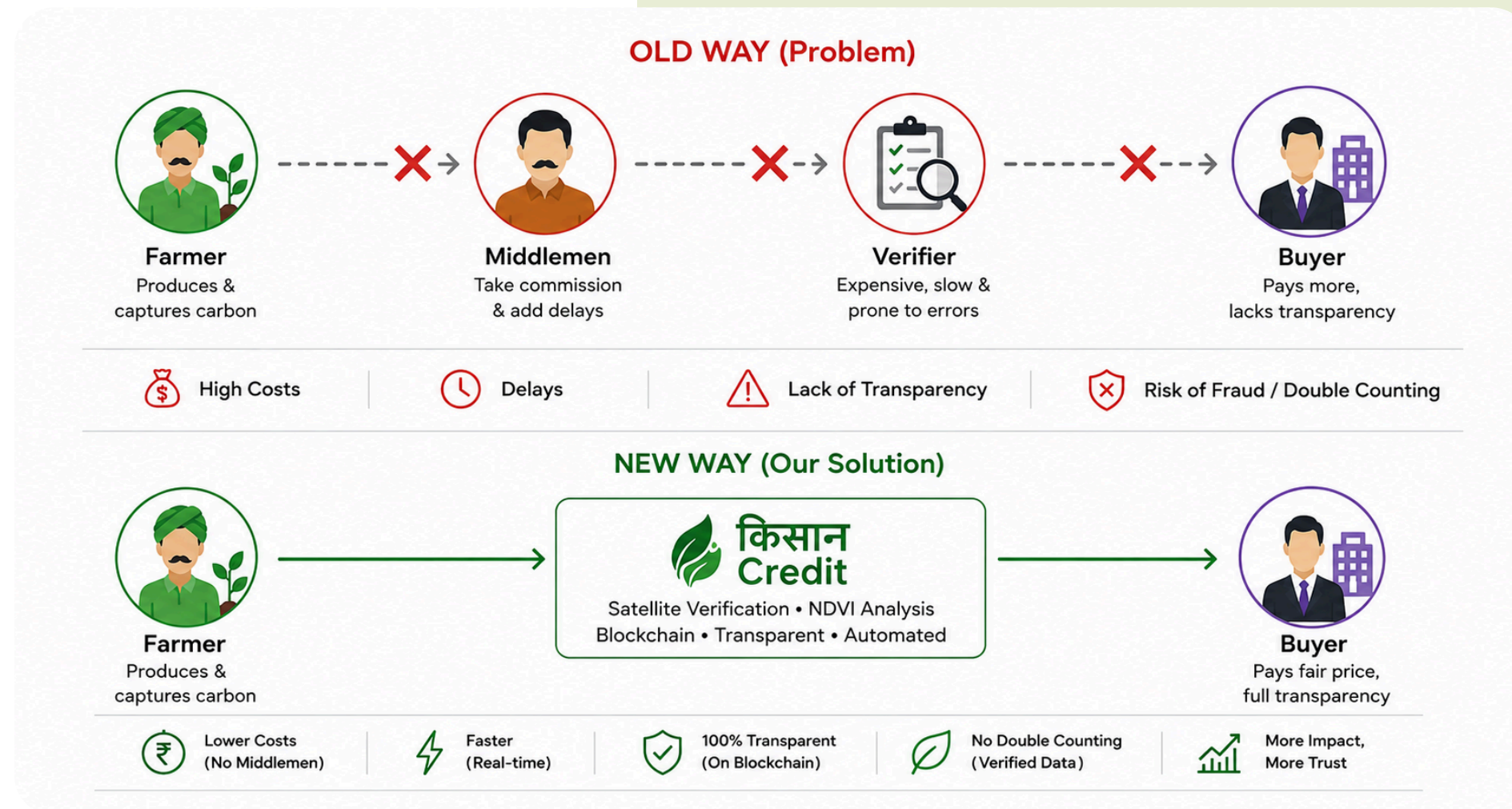
The Problem : Farmers are locked out of carbon money



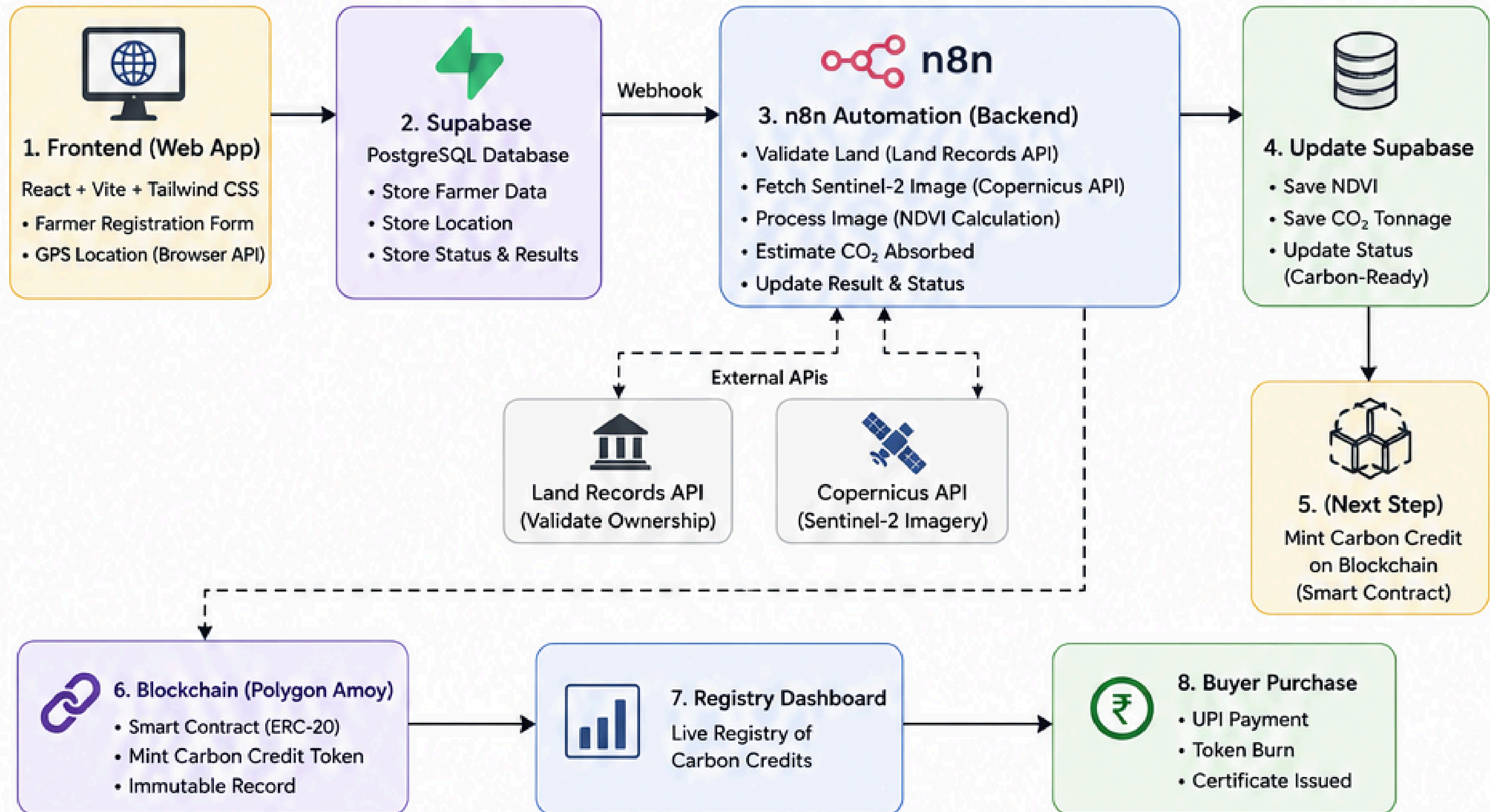
Core Insight of our Solution

Our insight: satellites can be the new “inspection officer”

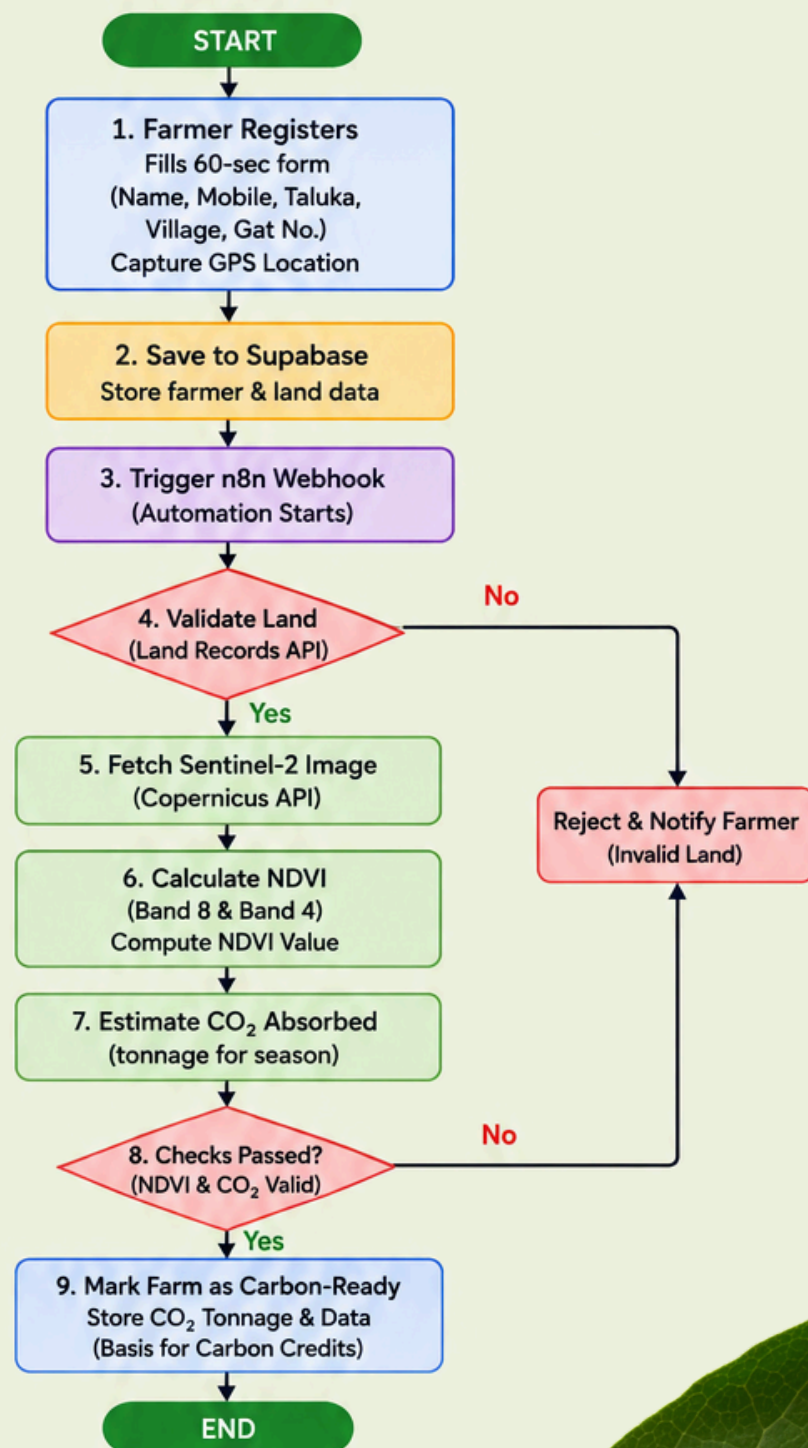
Free Sentinel-2 satellites already scan Nagpur farms every 5–12 days. NDVI (Normalized Difference Vegetation Index) tells us how green and healthy the crops are over time. Greener, healthier crops and better soil cover → more biomass → more carbon stored.



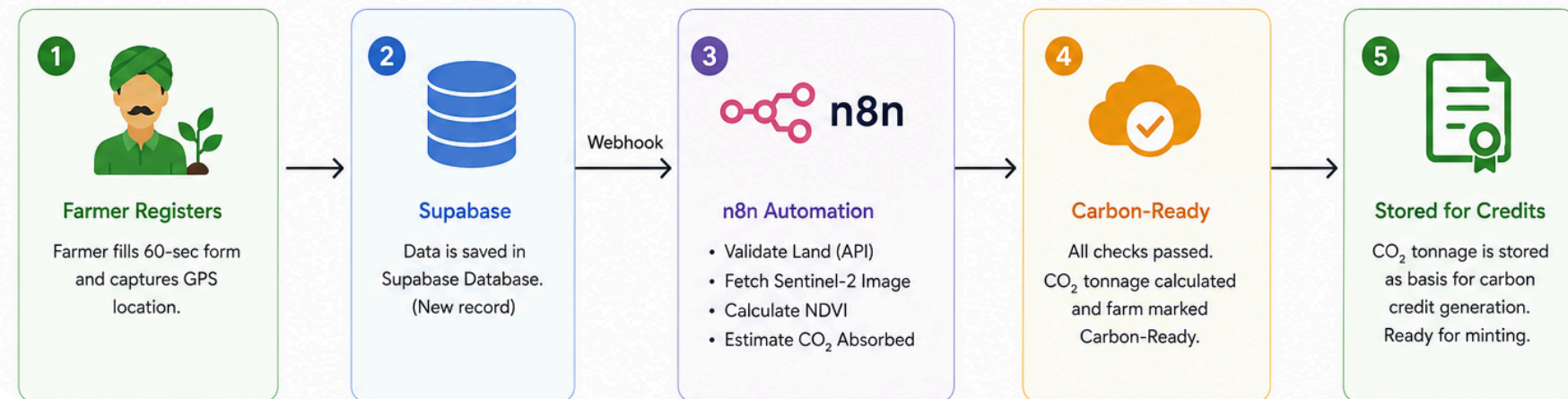
System Architecture



System Workflow



WORKFLOW



The flowchart represents the end-to-end automated workflow of the किसान Credit platform, where farmer registration data is captured through the web application, stored in Supabase, processed through n8n automation, validated using land records and Sentinel-2 satellite imagery, and analyzed using NDVI calculations to estimate seasonal CO₂ absorption. After verification, the farm is marked as Carbon-Ready and the verified CO₂ tonnage is stored for blockchain-based carbon credit generation.

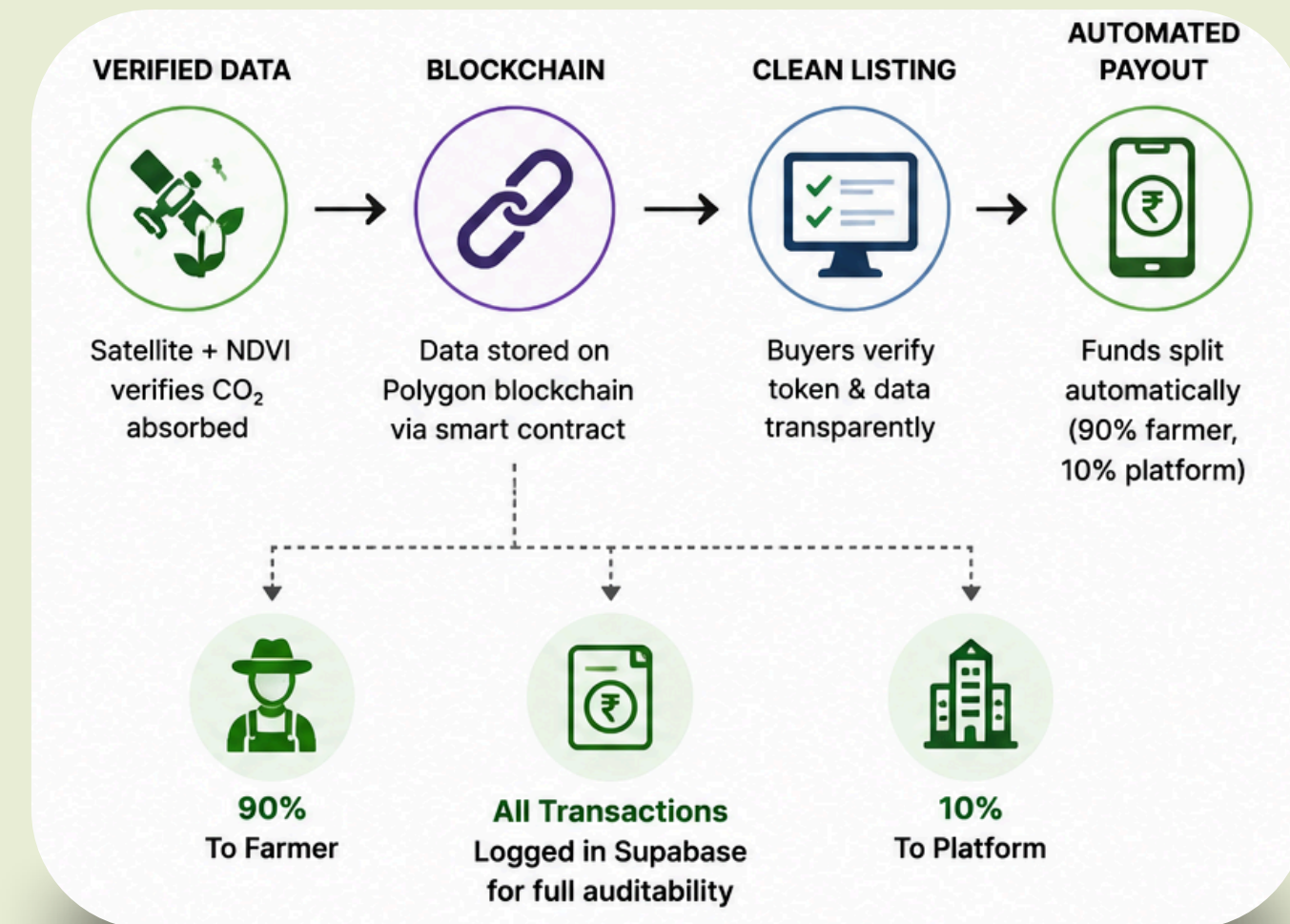
BLOCKCHAIN'S ROLE & SMART CONTRACTS

- Verified CO₂ data is pushed via ethers.js into a Solidity smart contract on Polygon, minting GEPCC carbon tokens mapped to that farm's ID, GPS and NDVI score, with a public Polygonscan transaction hash.
- Corporate buyers see a clean listing (e.g., "Nagpur orange farm 4.2 tons CO₂") and can click the hash to independently verify the token and its satellite trail, reducing greenwashing risk.
- On purchase (UPI/crypto in sandbox), payout logic automatically splits funds (e.g., 90% to farmer, 10% to platform) and logs every rupee and token movement into Supabase payments/transaction tables for full auditability.

"Code enforces who gets paid, blockchain enforces honesty, database enforces traceability."

From Data to Money: Trust & Payout

From Verified Data to Direct Payouts : No Middlemen



Thank

YOU

